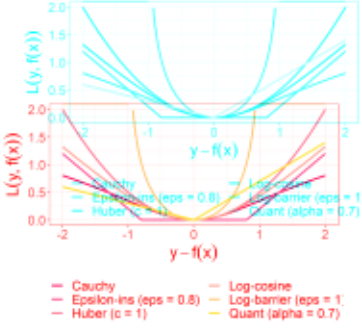
Advanced Risk Minimization

Advanced Regression Losses



Learning goals

- Know the Huber loss
- Know the log-cosh loss
- Know the Cauchy loss
- Know the log-barrier loss
- Know the ϵ-insensitive loss
- Know the quantile loss

Learning goals

- Huber loss
- Log-cosh loss
- Cauchy loss
- ϵ-insensitive loss
- Quantile loss

ADVANCED LOSS FUNCTIONS

Wang et al. 2020

Special loss functions can be used to estimate non-standard posterior components, to measure errors in a custom way or are designed to have special properties like robustness.

- Can use other losses for other posteriors
- Handle errors in a custom fashions

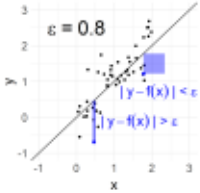
Examples:

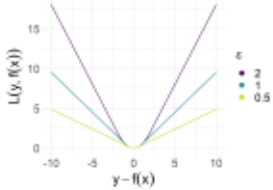
- Achieve properties like robustness
- Quantile loss: Overestimating a clinical parameter might not be as bad as underestimating it.
- Handle new predictive tasks
- Log-barrier loss: Extremely under- or overestimating demand in production would put company profit at risk.
- ϵ-insensitive loss: A certain amount of deviation in production does no harm, larger deviations do.

HUBER LOSS

$$L(y, f) = \begin{cases} \frac{1}{2}(y-f)^2 & \text{if } |y-f| \le \epsilon \\ \epsilon|y-f| - \frac{1}{2}\epsilon^2 & \text{otherwise} \end{cases}, \quad \epsilon > 0$$

- Piece-wise combination of $L1/L2$ to have robustness/smoothness
- Analytic properties: convex, differentiable (once)





- No closed-form solution even for constant or linear model.
- Solution behaves like **trimmed mean**: a (conditional) mean of two (conditional) quantiles.

No closed-form solution even for constant or linear model. No closed-form solution. To fit a model numerical optimization is necessary.
Solution behaves like **trimmed mean**: a (conditional) mean of two (conditional) quantiles.

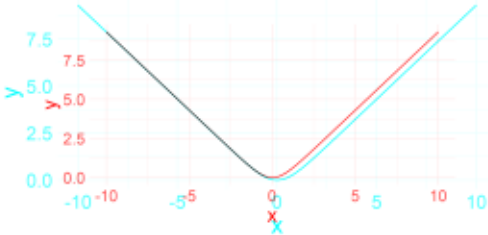
LOG-COSH LOSS

R. A. Saleh and A. Saleh 2022

$$L(y, f) = \log(\cosh(|y - f|)) \quad \text{where} \cosh(x) := \frac{e^x + e^{-x}}{2}$$

- Logarithm of the hyperbolic cosine of the residual.
- Approximately equal to $0.5(|y - f|)^2$ for small f and to $|y - f| - \log 2$ for large f, meaning it works mostly like $L2$ loss but is less outlier-sensitive.
- Has all the advantages of Huber loss and is, moreover, twice differentiable everywhere.

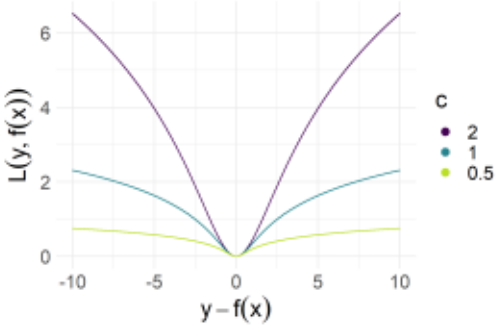
- Logarithm of the hyperbolic cosine of the residual.
- Approximately equal to $0.5(|y - f|)^2$ for small residuals and to $|y - f| - \log 2$ for large residuals, meaning it works a smoothed out $L1$ loss using $L2$ around the origin.
- Has all the advantages of Huber loss and is, moreover, twice differentiable everywhere.



CAUCHY LOSS

$$L(y, f) = \frac{c^2}{2} \log\left(1 + \left(\frac{|y - f|}{c}\right)^2\right), \quad c \in \mathbb{R}$$

- Particularly robust toward outliers (controllable via c).
- Analytic properties: differentiable, but not convex!

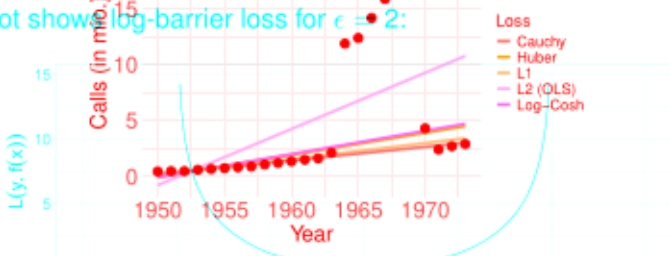


TELEPHONE DATA

We now illustrate the effect of using robust loss functions. The telephone data set contains the number of calls (in 10mio units) made in Belgium between 1950 and 1973 ($n = 24$). Outliers are due to a change in measurement without re-calibration for 6 years.

$$L(y, f) = \begin{cases} \epsilon^2 \log\left(\frac{(y-f)^2}{\epsilon^2}\right) & \text{if } |y-f| \leq \epsilon \\ (y-f)^2 & \text{if } |y-f| > \epsilon \end{cases}$$

- Behaves like $L2$ loss for small residuals.
- We use this if we don't want residuals larger than ϵ at all.
- No guarantee that the risk minimization problem has a solution.
- Plot shows log-barrier loss for $\epsilon = 2$:

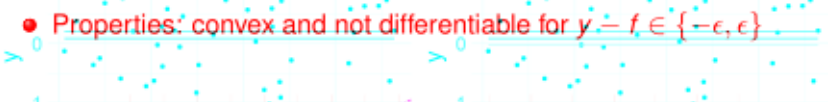


ϵ-INSENSITIVE LOSS

- Note that the optimization problem has no (finite) solution if there is no way to fit a constant where all residuals are smaller than ϵ.

$$L(y, f) = \begin{cases} 0 & \text{if } |y - f| \leq \epsilon \\ |y - f| - \epsilon & \text{otherwise} \end{cases} \quad \epsilon \in \mathbb{R}_+$$

- Modification of $L1$ loss, errors below ϵ accepted without penalty
- Used in SVM regression
- Properties: convex and not differentiable for $y - f \in \{-\epsilon, \epsilon\}$

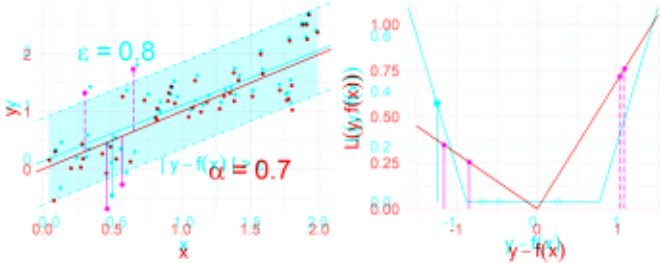


QUANTILE LOSS / PINBALL LOSS

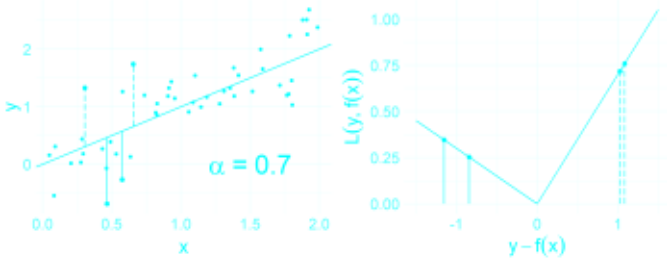
$$L(y,f) = \begin{cases} (1-\alpha)(f-y) & \text{if } y < f \\ \alpha(y-f) & \text{if } y \geq f \end{cases}, \quad \alpha \in (0,1)$$

$$L(y,f) = \begin{cases} 0 & \text{if } |y-f| \leq \epsilon \\ |y-f| - \epsilon & \text{otherwise} \end{cases}, \quad \epsilon \in \mathbb{R}_+$$

- Extension of $L1$ loss (equal to $L1$ for $\alpha = 0.5$).
- Modification of $L1$ loss, errors below ϵ accepted without penalty.
- Weighs either positive or negative residuals more strongly
- Used in SVM regression.
- $\alpha < 0.5$ ($\alpha > 0.5$) penalty to over-estimation (under-estimation)
- Properties: convex and not differentiable for $y - f \in \{-\epsilon, \epsilon\}$
- Risk minimizer is (conditional) α-quantile (median for $\alpha = 0.5$)



QUANTILE LOSS / PINBALL LOSS

$$L(y,f) = \begin{cases} (1-\alpha)(f-y) & \text{if } y < f \\ \alpha(y-f) & \text{if } y \geq f \end{cases}, \quad \alpha \in (0,1)$$

- Extension of $L1$ loss (equal to $L1$ for $\alpha = 0.5$).
- Weights either positive or negative residuals more strongly.
- $\alpha < 0.5$ ($\alpha > 0.5$) penalty to over-estimation (under-estimation)
- Risk minimizer is (conditional) α-quantile (median for $\alpha = 0.5$).